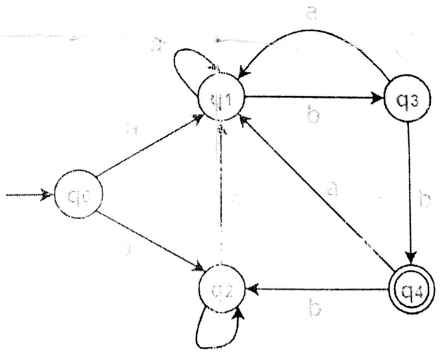
B. Tech. (Computer Engineering) Vth Semester
1st Sessional, AUTOMATA THEORY (CEN-502)

Time: 1 Hour MM:15

Note: Attempt all questions. Date: 18/09/23

Your Roll No : 21BCS028

Q No.	CO Mapped
1	1
2	2
3	1
4	2

Q. No	Question	Marks
1.	Design a minimum DFA which generates binary strings ($\sum = \{0, 1\}$) whose last two symbols are always same. ***Verify your answer using suitable examples.***	3
2.	Which of the following regular expressions are equivalent? i. $(00)^*(\varepsilon+0)$ ii. $(00)^*$ iii. 0^* iv. $0(00)^*$ ***Verify your answer using suitable examples***	3
3.	Minimize the following DFA using the Matrix Method.  ***Verify your minimized answer using suitable examples.***	5
4	Using Pumping Lemma, comment on the Regularity/ Non Regularity of the following languages which generate strings ***s*** as follows : i) $L_1 = \{ s \in \{0, 1\}^* \mid n_0(s) = n_1(s), \mid$ here $n_0(s)$ & $n_1(s)$ refer to no. of 0s & 1s in ***s*** $\}$ ii) $L_2 = \{ s \in \{0, 1\}^* \mid n_0(s)$ is a 2 digit prime number, $\mid n_0(s)$ refers to no. of 0s in ***s*** $\}$	4